Patrick Iben

Applied mathematics, biostatistics and research engineering

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GitHub | Research and teaching | ORCID 0009-0005-2443-7973

Profile

Applied mathematician and biostatistician with seven years in clinical biostatistics and twelve years in research. Works to eliminate needless ambiguity, risk, and suffering in decisions that rest on data. Builds statistical frameworks, AI-evaluation tools and public-data systems with traceable inputs and testable assumptions. Independent work includes six research manuscripts, scientific Python software, wearable sensing and 32 public-data consoles. Seeking research-scientist, research-engineering, AI-evaluation and biostatistical methods roles.

Technical expertise

Methods: trial design, estimands, causal inference under interference (spillover and contamination), consilience testing across data sources, simulation, permutation inference, calibration, decision curves, PK/NCA, bioequivalence, safety review, CDISC SDTM/ADaM and statistical analysis plans.

Tools: R (pharmaverse and Shiny), Quarto, SAS, Python, SQL, JavaScript, Git, Docker and LaTeX. Data provenance, automated QC, browser testing, public API integration and numerical validation.

Professional experience

Biostatistician II | Syneos Health | Remote | December 2021 to October 2026

- Served as lead biostatistician from December 2022 on early-phase clinical pharmacology studies, including first-in-human, SAD/MAD, crossover, bioequivalence and bioavailability designs.
- Owned protocol statistical sections, randomization schedules, analysis plans and mock tables, listings and figures; created or reviewed CDISC analysis specifications.
- Performed PK/PD analyses, reviewed outputs and contributed to clinical study reports; reviewed interim study data, annotated CRFs and database designs against analysis requirements.

Associate Biostatistician | Emmes | Rockville, MD | September 2019 to December 2021

- Led and supported federally sponsored Phase 1 and Phase 3 opioid-use-disorder trials; wrote protocol and analysis-plan sections, DSMB reports and final statistical reports.
- Built dynamic study-progress reporting and SAS/R tools; harmonized multi-clinic EHR data and supported external research partners.

Research Assistant and Teaching Assistant | University of Colorado Anschutz | August 2016 to July 2019

- Harmonized hospital databases containing millions of records and hundreds of derived variables for pediatric nursing-cost research and reuse in later studies.
- Advised doctoral nursing candidates on design and sample size; taught graduate biostatistics review sessions and office hours.

Earlier research roles | 2014 to 2016

Postsecondary and Policy Analysis Fellow, State Higher Education Executive Officers; Data Analysis Fellow, Yale School of Child Development; Senior Data Analyst, SCSU Office of Assessment and Planning.

Education

MS, Biostatistics, Data Science Analytics minor | University of Colorado Anschutz | 2019

BS, Mathematics, Applications of Mathematics | Southern Connecticut State University | 2016 | Magna cum laude

Selected projects

Trial biostatistics toolkit and eCRF review apps (worked examples, templates and apps for lead biostatisticians; example data synthetic)

- Built a worked bioequivalence study (2x2, N = 40) from sample size through SAP, ADaM, NCA, ICH M13A analysis, TLFs and CSR text in one script, with executed same-author QC (not independent double programming).

- Implemented safety-review screening rules (potential Hy's Law, ALT/AST confirmation flag, ICH E14 QTcF categories) and built a Safety Review Committee packet and in-browser Shiny console for a synthetic first-in-human study (654 expectations in 69 tests pass; not validated).
- Built a ten-gate QC harness with same-author recomputation and 18 pre-registered seeded defects (14 caught, 1 survived; not a detection rate); human sign-off pending.
- Developed 221 ADaM-based TLF pseudocode programs per language in SAS and R (SAS not executed; R parse-checked) and exploratory simulations of ICH M13A residual carryover and a C-QTc hysteresis check.
- Built four Python data-review apps (coding QC, de-identification, participant risk, central statistical monitoring) that run in the browser; published measured failures (the stand-in de-identifier left 108 of 308 identifiers readable) and a pre-registered check that routed sibling terms to human coding.

Public Data Consoles and Coastal Water Resilience

- Built 32 browser consoles integrating public observations, official forecasts and dated reference data, with source provenance, refresh checks, mobile layouts and no visitor API-key requirement.
- Adapted a ten-parameter research framework into data-informed response plans with action priorities, timing, escalation and parameter explanations. Coastal joins public station inventories, terrain, evacuation zones and NOAA feeds for geographic exposure screening; unsupported hydraulic quantities remain explicit.

Autonomic Telemetry and Resilience Telemetry

- Built Bluetooth-to-Mac-to-mobile wearable data transport with personal references, signal-quality and freshness controls, and replayable numerical views.
- Implemented cross-asset volatility-network diagnostics, historical replay and held-out comparisons against highest-volatility and persistence baselines as an exploratory adaptation of directed-contagion research.

AI evaluation and scientific software

- Implemented deterministic numeric and phrase checks, tool-ledger provenance, lexical retrieval evaluation and a containerized FastAPI service, with structured reports and executable known-gap tests.
- Evaluated calibration and decision curves against a synthetic oracle; packaged the Consilience Engine as an offline-capable Python library and configurable CLI.

Research and teaching

Six manuscripts have public drafts and paired concise and complete teaching editions. Research topics and review status as of 26 October 2026:

- **Cluster randomized trial design under spatial interference.** Monte Carlo evaluation of contamination control. *Statistics in Medicine*, submitted.
- **AlphaGenome within-site audit.** Permutation tests and replication across 11 genes. *Human Genomics*, submitted.
- **A False Floor.** Stronger-baseline counterexamples, training-catalog overlap and literature audit. PLOS ONE, submitted 22 October 2026.
- **Directed network contagion.** Directional asymmetry and simulated interventions across seven datasets. *Applied Network Science*, with editor.
- **Consilience Engine.** Permutation-based comparison of gene-group coherence across data modalities. *PeerJ*, in peer review.
- **Amygdala invariance.** Encoder-specific prediction and corroboration across three datasets. Peer Community In, preprint review.

Authored twelve educational editions with worked examples, answered exercises and manuscript-version provenance. Manuscripts, repositories, demonstrations and companions.